

Script Languages Final Project – Requirements Definition

1. General Purpose of the project.

The purpose of this project is to showcase the skills we have learnt during the Script Languages laboratories. It should include the graphical or web interface, and processes data in an organized way e.g. through the use of a database such as PostgreSQL. We also want to display our teamwork, since it has been growing across the laboratories. Thus the idea of the “Letterboxdle” (working title) was born.

The app is inspired by the popular web application Wordle, but instead of guessing words, the user would guess a movie title based on the reviews and distorted posters. This project should allow our team to display our best craftsmanship and also provide fun while coding it.

2. Problem statement

Movie-enjoyers often like to flex their knowledge and how niche there are. However forums often used by them often lack such a possibility. The application will give them an opportunity to showcase just how much they know about film, while drinking their flat white or matcha latte.

3. End-users:

- Performative Males – the team of python developers by that name, responsible for the design, functionality, and testing of the application
- The Course Instructor - evaluates the project based on technical functionality, error handling, code readability, database integration, **UI** aesthetics, and creativity.
- Performative males - the target audience of the application, players who will interact with the frontend interface

4. List of requirements:

a) (Basic) Functional Requirements:

- The program must include custom functions, other than GUI
- The program must rely on classes (object-oriented design)
- The program must implement input/output file handling (e.g., CSV, JSON, TXT)
- The program must use at least one external library
- The system must allow users to register a new account, log in securely, update their user profile details, and delete their account.
- The system must generate a game session consisting of a set number of rounds (e.g., 10 movies per game).
- The system should provide an endless game mode, where the player receives points for the streak of correct guesses.
- The system must dynamically adjust the blur/clarity level of the movie poster asset based on the current attempt using backend Python logic.
- The system must calculate points based on time remaining and the number of attempts used, updating the user's high score and statistics in the database.
- The system must allow authenticated users to export their gameplay history or match summary as a downloadable file
- For each movie, the user has three attempts to guess the title. Each incorrect guess must unlock a new hint:
 - *Attempt 1:* Displays a 5-star user review and a heavily blurred movie poster.
 - *Attempt 2:* Displays a 3-star user review and a slightly less blurred movie poster.

- Attempt 3: Displays a 1-star user review and an even less blurred poster.

b) Non-functional Requirements:

- The program must be developed using Python 3.8+
- The interface must be based on either Tkinter, Flask or Django
- The program must comply with PEP8 guidelines
- The project must include documentation (README.md)
- The program must be delivered as a GitHub repository (or a ZIP package)
- The project must be later presented during an in-person meeting in the classroom